🏆 Infosys SE Elite Crack Vault — 08 College ITM Pseudocode 35Q Exam Compendium Elite

🎯 Infosys Systems Engineer (SE) | 📊 24 Questions | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Problem 1 • Infosys SE Question 1

🔥 MUST-SOLVE

`ID: Q1` `Source: Infosys PYQ` `Level: Infosys SE Standard`

(Infosys System Engineer ITM Mock Paper 1 - Q46)
Section: Section 4: Pseudo Code | **Page:** 5
Visual Reference Plate: paper_01_page_05.png

Pseudocode / Algorithm Stem:

```
What is printed?
int f(int n) {
    if (n ⩽ 1) return n;
    return f(n-1) + 2 * f(n-2);
}
print(f(6))
```

A 15

B 21

C 27

D 31

✓ **Verified Answer:** (B)** — `21` **Step-by-Step Dry-Run Execution Trace & Analysis:** Recursive function trace: - f(0) = 0 - f(1) = 1 - f(2) = f(1) + 2·f(0) = 1 + 0 = 1 - f(3) = f(2) + 2·f(1) = 1 + 2(1) = 3 - f(4) = f(3) + 2·f(2) = 3 + 2(1) = 5 - f(5) = f(4) + 2·f(3) = 5 + 2(3) = 11 - f(6) = f(5) + 2·f(4) = 11 + 2(5) = 21.

Problem 2 • Infosys SE Question 2

MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q47)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_01_page_06.png

Pseudocode / Algorithm Stem:

```
What is printed?
x = 45; count = 0;
while (x ≠ 0) {
    x = x & (x - 1);
    count++;
}
print(count ^ (45 >> 2))
```

A 4

B 11

C 15

D 7

✓ **Verified Answer:** (C)** — `15` **Step-by-Step Dry-Run Execution Trace & Analysis:** Brian Kernighan's bit-counting loop clears the lowest set bit in each iteration. For x = 45 = 32 + 8 + 4 + 1 = 101101_2 (4 set bits): Loop executes 4 times ⇒ count = 4. Right shift: 45 >> 2 = 11 (binary 1011_2). Bitwise XOR: count ^ 11 = 4 ^ 11 = 0100_2 ^ 1011_2 = 1111_2 = 15.

Problem 3 • Infosys SE Question 3

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q48)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_01_page_06.png

Pseudocode / Algorithm Stem:

```
What is printed?
s = 0
for i = 1 to 5
    for j = i to 5
        if ((i + j) % 3 == 0) continue
        if (j == 5) break
        s = s + i * j
    end for
end for
print(s)
```

A 38

B 42

C 46

D 50

✓ **Verified Answer:** (C)** — `46` **Step-by-Step Dry-Run Execution Trace & Analysis:** Nested loops simulation: - i=1: j=1 (sum=2, add 1); j=2 ((1+2)%3==0, continue); j=3 (sum=4, add 3); j=4 (sum=5, add 4); j=5 ((1+5)%3==0, continue before break!). Running s = 1 + 3 + 4 = 8. - i=2: j=2 (sum=4, add 4); j=3 (sum=5, add 6); j=4 ((2+4)%3==0, continue); j=5 (j==5, break). Running s = 8 + 4 + 6 = 18. - i=3: j=3 ((3+3)%3==0, continue); j=4 (sum=7, add 12); j=5 (j==5, break). Running s = 18 + 12 = 30. - i=4: j=4 (sum=8, add 16); j=5 ((4+5)%3==0, continue before break). Running s = 30 + 16 = 46. - i=5: j=5 (j==5, break). Final s = 46.

Problem 4 • Infosys SE Question 4

MUST-SOLVE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q49)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_01_page_06.png

Pseudocode / Algorithm Stem:

```
A stack is used (operands are evaluated left to right). What is printed?
push(3); push(5); push(7);
x = pop();
push(x * 2);
push(pop() + pop());
print(pop() - pop())
```

A 14

B 16

C 19

D 22

✓ **Verified Answer:** (B)** — `16` **Step-by-Step Dry-Run Execution Trace & Analysis:** Stack execution trace (LIFO): 1) push(3) → [3] 2) push(5) → [3, 5] 3) push(7) → [3, 5, 7] 4) x = pop() ⇒ x = 7; stack = [3, 5] 5) push(7 * 2) = push(14) → [3, 5, 14] 6) push(pop() + pop()): first pop() returns 14, second pop() returns 5; 14 + 5 = 19; push(19) → [3, 19] 7) print(pop() - pop()): first pop() returns 19, second pop() returns 3; 19 - 3 = 16.

Problem 5 • Infosys SE Question 5

MUST-SOLVE

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q50)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_01_page_06.png

Pseudocode / Algorithm Stem:

```
What is the final content of arr? (indexing starts at 0; updates are in place)
arr = [4, 1, 3, 2]
for i = 0 to 3
    arr[i] = arr[i] + arr[arr[i] % 4]
end for
```

A [8, 2, 5, 7]

B [8, 2, 5, 6]

C [8, 5, 5, 7]

D [5, 2, 5, 7]

✓ **Verified Answer:** (A)** — `[8, 2, 5, 7]` **Step-by-Step Dry-Run Execution Trace & Analysis:** In-place array update with arr = [4, 1, 3, 2]: - i = 0: arr[0] = arr[0] + arr[arr[0] % 4] = 4 + arr[4 % 4] = 4 + arr[0] = 4 + 4 = 8. arr = [8, 1, 3, 2]. - i = 1: arr[1] = arr[1] + arr[arr[1] % 4] = 1 + arr[1 % 4] = 1 + arr[1] = 1 + 1 = 2. arr = [8, 2, 3, 2]. - i = 2: arr[2] = arr[2] + arr[arr[2] % 4] = 3 + arr[3 % 4] = 3 + arr[3] = 3 + 2 = 5. arr = [8, 2, 5, 2]. - i = 3: arr[3] = arr[3] + arr[arr[3] % 4] = 2 + arr[2 % 4] = 2 + arr[2] = 2 + 5 = 7. arr = [8, 2, 5, 7]. Final content: [8, 2, 5, 7].

Problem 6 • Infosys SE Question 6

MUST-SOLVE

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q46)
Section: Section 4: Pseudo Code | **Page:** 5
Visual Reference Plate: paper_02_page_05.png

Pseudocode / Algorithm Stem:

```
What is printed?
int f(int n) {
    if (n == 0) return 1;
    if (n == 1) return 2;
    return f(n-1) + f(n-2) * 2;
}
print(f(5))
```

A 26

B 30

C 32

D 34

✓ **Verified Answer:** (C)** — `32` **Step-by-Step Dry-Run Execution Trace & Analysis:** 32. f(0)=1, f(1)=2, f(2)=2+2=4, f(3)=4+4=8, f(4)=8+8=16, f(5)=16+16=32.

Problem 7 • Infosys SE Question 7

MUST-SOLVE

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q47)
Section: Section 4: Pseudo Code | **Page:** 5
Visual Reference Plate: paper_02_page_05.png

Pseudocode / Algorithm Stem:

```
What is printed? (x and y are unsigned integers)
x = 58; y = 0;
while (x > 0) {
    y = (y << 1) | (x & 1);
    x = x >> 1;
}
print(y)
```

A 23

B 29

C 37

D 41

✓ **Verified Answer:** (A)** — `23` **Step-by-Step Dry-Run Execution Trace & Analysis:** 23. The loop peels off bits of x from the LSB and rebuilds y by shifting left each time, which reverses the bit order of x within its significant bits. 58 = 111010_2 (6 bits); reversed = 010111_2 = 23. (Trace: y goes 0→0→1→2→5→11→23 as x's bits 0,1,0,1,1,1 are consumed in order.)

Problem 8 • Infosys SE Question 8

MUST-SOLVE

ID: Q8 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q48)
Section: Section 4: Pseudo Code | **Page:** 5
Visual Reference Plate: paper_02_page_05.png

Pseudocode / Algorithm Stem:

```
What is printed?
s = 0
for i = 1 to 4
    for j = 1 to 4
        if (i == j) continue
        if (i + j > 6) break
        s = s + (i * j)
    end for
end for
print(s)
```

A 40

B 44

C 46

D 48

✓ **Verified Answer:** (C)** — `46` **Step-by-Step Dry-Run Execution Trace & Analysis:** 46. i=1: skips j=1 (i==j); adds 1×2+1×3+1×4 = 9. i=2: skips j=2; adds 2×1+2×3+2×4 = 16 (running 25; note j=4 gives i+j=6, which is not >6, so it's included). i=3: skips j=3; adds 3×1+3×2 = 9, then j=4 gives i+j=7>6 → break (running 34). i=4: adds 4×1+4×2 = 12, then j=3 gives i+j=7>6 → break (running 46).

Problem 9 • Infosys SE Question 9

CORE SE

ID: Q9 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q49)
Section: Section 4: Pseudo Code | **Page:** 5
Visual Reference Plate: paper_02_page_05.png

Pseudocode / Algorithm Stem:

```
A queue is used (FIFO). What is printed?
enqueue(2); enqueue(4); enqueue(6);
a = dequeue();
enqueue(a + 10);
b = dequeue();
c = dequeue();
print(b + c)
```

A 10

B 16

C 18

D 22

✓ **Verified Answer:** (A)** — `10` **Step-by-Step Dry-Run Execution Trace & Analysis:** 10. Queue [2,4,6] → dequeue 2 (a) → [4,6] → enqueue 12 → [4,6,12] → dequeue 4 (b) → dequeue 6 (c) → [12]. b + c = 4 + 6 = 10.

Problem 10 • Infosys SE Question 10

CORE SE

ID: Q10 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q50)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_02_page_06.png

Pseudocode / Algorithm Stem:

```
What is the final content of arr? (0-indexed; each statement uses the array's state at that point)
arr = [1, 2, 3, 4, 5]
temp = arr[0]
arr[0] = arr[4]
arr[4] = arr[2]
arr[2] = arr[1]
arr[1] = temp
```

A [5, 1, 2, 4, 3]

B [5, 1, 3, 4, 2]

C [5, 2, 1, 4, 3]

D [5, 1, 2, 4, 2]

✓ **Verified Answer:** (A)** — `[5, 1, 2, 4, 3]` **Step-by-Step Dry-Run Execution Trace & Analysis:** temp=1. arr[0]=arr[4]=5 → [5,2,3,4,5]. arr[4]=arr[2]=3 → [5,2,3,4,3]. arr[2]=arr[1]=2 → [5,2,2,4,3]. arr[1]=temp=1 → final [5,1,2,4,3].

Problem 11 • Infosys SE Question 11

CORE SE

ID: Q11 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q46)
Section: Section 4: Pseudo Code | **Page:** 5
Visual Reference Plate: paper_03_page_05.png

Pseudocode / Algorithm Stem:

```
What is printed?
```

A 40

B 46

C 52

D 44

1 + f(n-2) + n;

✓ **Verified Answer:** (B)** — `46` **Step-by-Step Dry-Run Execution Trace & Analysis:** 46. f(0)=0, f(1)=1, f(2)=1+0+2=3, f(3)=3+1+3=7, f(4)=7+3+4=14, f(5)=14+7+5=26, f(6)=26+14+6=46.

Problem 12 • Infosys SE Question 12

CORE SE

ID: Q12 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q47)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_03_page_06.png

Pseudocode / Algorithm Stem:

```
What is printed?
```

A 37

B 41

C 45

D 39

✓ **Verified Answer:** (B)** — `41` **Step-by-Step Dry-Run Execution Trace & Analysis:** 41. 37 = 100101_2 (bits: bit0=1,bit2=1,bit5=1, rest 0). The loop places bit i of x into position (5−i) of mask, which reverses the 6-bit pattern: 100101 reversed is 101001 = 32+8+1 = 41.

Problem 13 • Infosys SE Question 13

CORE SE

ID: Q13 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q48)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_03_page_06.png

Pseudocode / Algorithm Stem:

```
What is printed?
```

A 130

B 124

C 136

D 118

✓ **Verified Answer:** (A)** — `130` **Step-by-Step Dry-Run Execution Trace & Analysis:** 130. Tracing directly (j increments before each continue, so the loop always terminates correctly): i=1 contributes 1; i=2 contributes 2; i=3 contributes 3+9=12; i=4 contributes 4+12=16; i=5 contributes 5+15+25=45; i=6 contributes 6+18+30=54. Total = 1+2+12+16+45+54 = 130.

Problem 14 • Infosys SE Question 14

CORE SE

ID: Q14 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q49)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_03_page_06.png

Pseudocode / Algorithm Stem:

```
A stack is used (LIFO). What is printed?
```

A 39

B 33

C 42

D 30

✓ **Verified Answer:** (A)** — `39` **Step-by-Step Dry-Run Execution Trace & Analysis:** 39. Stack: push 4,7 → [4,7]. a=pop=7, b=pop=4 → push(7−4=3) → [3]. push(10) → [3,10]. c=pop=10, d=pop=3 → push(13) → [13]. push(3) → [13,3]. e=pop=3, f=pop=13. e×f = 39.

Problem 15 • Infosys SE Question 15

CORE SE

ID: Q15 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q50)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_03_page_06.png

Pseudocode / Algorithm Stem:

```
What is the final content of arr? (0-indexed; each statement uses the array's state at that point)
```

A [60, 40, 30, 20, 50, 10]

B [60, 20, 30, 40, 50, 10]

C [10, 40, 30, 20, 50, 60]

D [60, 40, 50, 20, 30, 10]

✓ **Verified Answer:** (A)** — `[60, 40, 30, 20, 50, 10]` **Step-by-Step Dry-Run Execution Trace & Analysis:** i=0,j=5: swap arr[0],arr[5] → [60,20,30,40,50,10]; i=1,j=3. i

Problem 16 • Infosys SE Question 16

CORE SE

ID: Q16 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q46)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_04_page_06.png

Pseudocode / Algorithm Stem:

```
What is printed? (% is the modulo operator)
int gcd(int a, int b) {
    if (b == 0) return a;
    return gcd(b, a % b);
}
print(gcd(252, 105))
```

A 21

B 42

C 7

D 14

✓ **Verified Answer:** (A)** — `21` **Step-by-Step Dry-Run Execution Trace & Analysis:** 21. gcd(252,105): 252%105=42 → gcd(105,42): 105%42=21 → gcd(42,21): 42%21=0 → gcd(21,0)=21.

Problem 17 • Infosys SE Question 17

CORE SE

ID: Q17 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q47)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_04_page_06.png

Pseudocode / Algorithm Stem:

```
What is printed? (^=XOR, |=OR, &=AND, >>=right shift)
a = 13; b = 25;
c = (a ^ b) & (a | b);
d = c >> 2;
print(c + d)
```

A 25

B 29

C 20

D 33

✓ **Verified Answer:** (A)** — `25` **Step-by-Step Dry-Run Execution Trace & Analysis:** 25. 13=01101, 25=11001. a^b = 10100 = 20. a|b = 11101 = 29. c = 20 & 29 = 10100 = 20. d = 20>>2 = 5. c+d = 20+5 = 25.

Problem 18 • Infosys SE Question 18

CORE SE

ID: Q18 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q48)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_04_page_06.png

Pseudocode / Algorithm Stem:

```
How many swaps are performed in total?
arr = [5, 2, 8, 1, 9, 3]
swaps = 0
for i = 0 to 4
    for j = 0 to 4 - i
        if (arr[j] > arr[j+1])
            temp = arr[j]
            arr[j] = arr[j+1]
            arr[j+1] = temp
            swaps = swaps + 1
        end if
    end for
end for
print(swaps)
```

A 7

B 8

C 6

D 9

✓ **Verified Answer:** (A)** — `7` **Step-by-Step Dry-Run Execution Trace & Analysis:** 7. Full bubble-sort trace: pass i=0 gives 3 swaps (array becomes [2,5,1,8,3,9]); pass i=1 gives 2 more swaps ([2,1,5,3,8,9]); pass i=2 gives 2 more swaps ([1,2,3,5,8,9], now sorted); passes i=3 and i=4 give 0 swaps. Total = 3+2+2 = 7.

Problem 19 • Infosys SE Question 19

CORE SE

ID: Q19 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q49)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_04_page_06.png

Pseudocode / Algorithm Stem:

```
This algorithm reverses the first k elements of a queue using a stack. What is printed?
queue = [1, 2, 3, 4, 5]
stack = []
k = 3
for i = 1 to k
    push(stack, dequeue(queue))
end for
while (stack is not empty)
    enqueue(queue, pop(stack))
end while
for i = 1 to (5 - k)
    enqueue(queue, dequeue(queue))
end for
print(queue)
```

A [3, 2, 1, 4, 5]

B [1, 2, 3, 4, 5]

C [5, 4, 3, 2, 1]

D [3, 2, 1, 5, 4]

✓ **Verified Answer:** (A)** — `[3, 2, 1, 4, 5]` **Step-by-Step Dry-Run Execution Trace & Analysis:** This is the standard "reverse first k of a queue" algorithm. After moving the first 3 elements through the stack, the queue becomes [4,5,3,2,1]; the final rotation step (dequeue+enqueue twice) moves 4 and 5 to the back, giving [3,2,1,4,5] — the first 3 elements correctly reversed.

Problem 20 • Infosys SE Question 20

CORE SE

ID: Q20 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q50)
Section: Section 4: Pseudo Code | **Page:** 7
Visual Reference Plate: paper_04_page_07.png

Pseudocode / Algorithm Stem:

```
This algorithm moves all non-zero elements to the front, preserving order, and fills the rest with zero
What is the final arr?
arr = [3, 0, 4, 0, 5, 0, 2]
pos = 0
for i = 0 to 6
    if (arr[i] ≠ 0)
        arr[pos] = arr[i]
        pos = pos + 1
    end if
end for
while (pos ≤ 6)
    arr[pos] = 0
    pos = pos + 1
end while
print(arr)
```

A [3, 4, 5, 2, 0, 0, 0]

B [3, 0, 4, 0, 5, 0, 2]

C [3, 4, 5, 2, 0, 0, 2]

D [0, 0, 0, 3, 4, 5, 2]

✓ **Verified Answer:** (A)** — `[3, 4, 5, 2, 0, 0, 0]` **Step-by-Step Dry-Run Execution Trace & Analysis:** Non-zero elements (3,4,5,2) are copied to the front in order, giving positions 0–3; the while loop then zeroes out positions 4,5,6. Final: [3,4,5,2,0,0,0].

Problem 21 • Infosys SE Question 21

CORE SE

ID: Q21 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q46)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_05_page_06.png

Pseudocode / Algorithm Stem:

```
What is printed? (This computes the number of grid paths from one corner to another, moving only
right or down.)
int f(int n, int m) {
    if (n == 0 || m == 0) return 1;
    return f(n-1, m) + f(n, m-1);
}
print(f(3, 2))
```

A 10

B 8

C 12

D 9

✓ **Verified Answer:** (A)** — `10` **Step-by-Step Dry-Run Execution Trace & Analysis:** 10. This recursion counts grid paths and equals C(n+m,n). Tracing directly: f(1,1)=2, f(2,1)=3, f(1,2)=3, f(2,2)=6, f(3,1)=4, f(3,2)=f(2,2)+f(3,1)=6+4=10 (also confirmed by code execution).

Problem 22 • Infosys SE Question 22

CORE SE

ID: Q22 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q47)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_05_page_06.png

Pseudocode / Algorithm Stem:

```
What is printed? (⟫=right shift, ⟪=left shift, &=AND, ^=XOR)
x = 0;
for i = 0 to 7
    if (((45 >> i) & 1) == 1)
        x = x + (1 << (7 - i))
    end if
end for
y = x ^ 45
print(y)
```

A 153

B 180

C 135

D 171

✓ **Verified Answer:** (A)** — `153` **Step-by-Step Dry-Run Execution Trace & Analysis:** 153. 45 = 00101101 in 8 bits. The loop reverses these 8 bits into x, giving x=180 (10110100). y = x^45 = 180^45 = 153 (verified by direct computation).

Problem 23 • Infosys SE Question 23

CORE SE

ID: Q23 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q48)
Section: Section 4: Pseudo Code | **Page:** 6
Visual Reference Plate: paper_05_page_06.png

Pseudocode / Algorithm Stem:

```
This is a binary search for target = 50 in a sorted array. How many comparisons are made before
the loop ends (the target is not present)?
arr = [2, 5, 8, 12, 16, 23, 38, 45, 56, 72, 91]
low = 0; high = 10; target = 50; comparisons = 0
while (low ⩽ high)
    mid = (low + high) / 2
    comparisons = comparisons + 1
    if (arr[mid] == target)
        break
    else if (arr[mid] < target)
        low = mid + 1
    else
        high = mid - 1
    end if
end while
print(comparisons)
```

A 4

B 3

C 5

D 6

✓ **Verified Answer:** (A)** — `4` **Step-by-Step Dry-Run Execution Trace & Analysis:** 4. Tracing the search for target=50 (not present): mid=5(23)→low=6; mid=8(56)→high=7; mid=6(38)→low=7; mid=7(45)→low=8, loop ends. That's 4 comparisons (verified by direct trace).

Problem 24 • Infosys SE Question 24

CORE SE

ID: Q24 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q49)
Section: Section 4: Pseudo Code | **Page:** 7
Visual Reference Plate: paper_05_page_07.png

Pseudocode / Algorithm Stem:

```
This stack trace evaluates a postfix-style expression. What is printed?
push(6); push(2); push(3);
c = pop(); b = pop(); push(b + c);
d = pop(); e = pop(); push(e - d);
push(3); push(8); push(2);
f = pop(); g = pop(); push(g / f);
h = pop(); i = pop(); push(i + h);
j = pop(); k = pop(); push(k * j);
print(pop())
```

A 7

B 10

C 14

D 4

✓ **Verified Answer:** (A)** — `7` **Step-by-Step Dry-Run Execution Trace & Analysis:** 7. This performs the postfix evaluation of "6 2 3 + − 3 8 2 / + ×": (2+3)=5, (6−5)=1, (8/2)=4, (3+4)=7, (1×7)=7.